

Tutorial 1: Number System and Data Representation

Question 1

Fill in the blanks with equivalent value for each numbering system:

Decimal	Binary	Octal	Hexadecimal	Gray
237 ₁₀	1110 1101 ₂	355 ₈	ED ₁₆	1001 1011 _{Gray}
22 ₁₀	0001 0110 ₂	26 ₈	16 ₁₆	0001 1101 _{Gray}
252 ₁₀	1111 1100 ₂	374 ₈	FC ₁₆	1000 0010 _{Gray}
241 ₁₀	1111 0001 ₂	361 ₈	F1 ₁₆	1000 1001 _{Gray}
26 ₁₀	0001 1010 ₂	32 ₈	1A ₁₆	10111 _{Gray}

Question 2

Convert the Binary numbers to BCD numbers (and vice versa)

Binary	Dec	BCD
0100 1001 ₂	73 ₁₀	0111 0011 _{BCD}
0110 0000 ₂	96 ₁₀	1001 0110 _{BCD}
1011 0001 ₂	177 ₁₀	1 0111 0111 _{BCD}
Invalid		1101 0101 _{BCD}

Question 3

Perform the following calculation, assuming an 8-bit system.

- (i) Calculate the result of $37_8 + 100101_2$ and give the answer in BCD

$$\begin{aligned} 37_8 + 100101_2 &= 011\ 111_2 + 100101_2 \\ &= 100\ 0100_2 \\ &= (64 + 4)_{10} \\ &= 68_{10} \\ &= 0110\ 1000_{\text{BCD}} \end{aligned}$$

- (ii) Calculate the result of $0110\ 0101_2 - 0100\ 0000_2$ and leave the answer in 8-bit 2's complement

$$\begin{aligned} &0110\ 0101_2 - 0100\ 0000_2 \\ &= 0110\ 0101_2 + (-0100\ 0000_2) \\ &= 0110\ 0101_2 + 1100\ 0000_2 \quad (2\text{'s complement}) \\ &= 0010\ 0101_2 \end{aligned}$$

(iii) Calculate $2A_{16} - 147_8$ and give the answer in binary

$$\begin{aligned} 2A_{16} - 147_8 &= 0010\ 1010_2 - 01\ 100\ 111_2 \\ &= 0010\ 1010_2 + 1001\ 1001_2 \\ &= 1100\ 0011_2 \end{aligned}$$

	0110 0111 ₂
1's complement:	1001 1000 ₂
2's complement:	1001 1001 ₂

(iv) Calculate $0111\ 1101_{\text{Gray}} - 0101\ 1110_2$ and give the answer in binary

$$\begin{aligned} 0111\ 1101_{\text{Gray}} - 0101\ 1110_2 \\ &= 0101\ 0110_2 + 1010\ 0010_2 \\ &= 1111\ 1000_2 \end{aligned}$$

	0101 1110 ₂
1's complement:	1010 0001 ₂
2's complement:	1010 0010 ₂